Institute of Technology of Cambodia

Department of Information and Communication Engineering

Assignment: TP09

Subject: **Operating System**

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Objectives

Create a bash script to perform user management tasks as outlined below:

- Create a new group. Each group must have a unique name. The script must check to ensure that no duplicate group names exist on the system. If a duplicate is found, an error needs to be reported, and the administrator must try another group name.
- Create a new user. Each user must have a unique name. The script must check to ensure that no duplicate usernames exist on the system. If a duplicate is found, an error needs to be reported and the administrator must try another username. The user will have a Bash login shell and belong to the group that was created in the previous step.
- Create a password for each user that is created.
- Ensure that the new user created is a member of the new group created.
- Create a directory at the root / of the file system with same name as the user created.
- Set the ownership of the directory to the user and group created.
- Set the permissions of the directory to full control for the owner and full control for the group created.
- Set the permissions to ensure that only the owner of a file can delete it from the directory.
- Ensure that the script is executable.

```
#!/bin/bash

#function for create group
create_group(){
	while true;do
		read -p "Enter group name: " group_name
		if [ $(getent group $group_name | wc -l) -ne 0 ]
		then
			echo "Group '$group_name' already exist. Please try other name."
		else
			sudo groupadd $group_name
			echo "Group '$group_name' created succcessfully!"
			break
		fi
	done
}
create_user(){
	while true;do
		read -p "Enter user name: " user_name
		if [ $(getent user $user_name | wc -l ) -ne 0]
		then
			echo "User '$user_name' already exist. Please try other name."
		else
			sudo useradd -m -s /bin/bash -g $group_name  $user_name
			echo "User'$user_name' created successfully!"
```

```
                    break
            fi
      done

}

set_password(){
      echo "Setting password for user '$user_name': "
      sudo passwd $user_name

}
create_user_directory(){
      dir="/$user_name"
      sudo mkdir "$dir"
      sudo chown "$user_name":"$group_name" "$dir"
      sudo chmod 1770 "$dir"
      echo "Directory '$user_name' created and permissons set."
}

create_group
create_user
set_password
create_user_directory
```

Result:

```
sokmeak@sokmeaksaren:~/bin$ user_management
Enter group name: Test3
Group 'Test3' created succcessfully!
Enter user name: dara
Unknown database: user
/home/sokmeak/bin/user_management: line 22: [: missing `]'
User'dara' created successfully!
Setting password for user 'dara':
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
Directory 'dara' created and permissons set.
sokmeak@sokmeaksaren:~/bin$ getent group dara
sokmeak@sokmeaksaren:~/bin$ id dara
uid=1008(dara) gid=1013(Test3) groups=1013(Test3)
sokmeak@sokmeaksaren:~/bin$ groups dara
dara : Test3
sokmeak@sokmeaksaren:~/bin$ cd /
sokmeak@sokmeaksaren:/$ ls -l | grep dara
drwxrwx--T   2 dara Test3       4096 Jan 10 07:24 dara
```

+ File organization: The script organizes files into predefined categories (e.g., `Images`, `Documents`, `Scripts`, `Archives`) based on their file extensions otherwise any file that doesn't match the specified extensions is placed in the `Others` category.

```
  GNU nano 7.2                                                    fileOrganizer *
#!/bin/bash

# Get the current directory
current=$(pwd)
cd "$current" || exit

# Define categories and associated extensions
declare -A categories=(
    [Images]="jpg png gif jpeg"
    [Documents]="docx txt pdf pptx xlsx"
    [Scripts]="sh jar c cpp py js"
    [Archives]="zip tar.gz rar"
    [Others]="*"
)

# Loop through files in the directory
for file in *; do
    # Skip directories
    if [ -d "$file" ]; then
        continue
    fi

    # Default category
    category="Others"

    # Determine the file extension
    extension="${file##*.}"

    # Check for the category of the file
    for cat in "${!categories[@]}"; do
        for ext in ${categories[$cat]}; do
            if [[ "$extension" == "$ext" ]]; then
                category="$cat"
                break 2
            fi
        done
    done

    # Create directory if it doesn't exist
    if [ ! -d "$category" ]; then
        mkdir "$category"
    fi

    # Move the file to the respective category directory
    mv "$file" "$category/"
    echo "Moved '$file' to '$category/'"
done

echo "Files are organized by categories."
```

```
sokmeak@sokmeaksaren:~/bin$ ls
checkFileExtension  eoneb  etwob  etwod          getStarted      myCalculation.docx  organizeMe  user_management
eonea               etwoa  etwoc  fileOrganizer  hello_world.sh  myNote.txt          OS.pptx     user_management_Script.tx
sokmeak@sokmeaksaren:~/bin$ fileOrganizer
Moved 'checkFileExtension' to 'Others/'
Moved 'eonea' to 'Others/'
Moved 'eoneb' to 'Others/'
Moved 'etwoa' to 'Others/'
Moved 'etwob' to 'Others/'
Moved 'etwoc' to 'Others/'
Moved 'etwod' to 'Others/'
Moved 'fileOrganizer' to 'Others/'
Moved 'getStarted' to 'Others/'
Moved 'hello_world.sh' to 'Scripts/'
Moved 'myCalculation.docx' to 'Documents/'
Moved 'myNote.txt' to 'Documents/'
Moved 'organizeMe' to 'Others/'
Moved 'OS.pptx' to 'Documents/'
Moved 'user_management' to 'Others/'
Moved 'user_management_Script.txt' to 'Documents/'
Files are organized by categories.
```

```
sokmeak@sokmeaksaren:~/bin$ ls
Documents  Others  Scripts
sokmeak@sokmeaksaren:~/bin$
```